



NUCLEAR ENERGY — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Tarapur — India's First Nuclear Power Station (1969)

- Tarapur Atomic Power Station (TAPS, Boisar, Thane district, Maharashtra) **India ka pehla nuclear power station hai — 1969 mein commercial operation shuru hua.** [Q1]

2. Nuclear Share in Total Energy (~1.77-4%)

- 2006-07 mein nuclear energy ka share **3-4% ke beech tha** — 31 Dec 2024 tak **nuclear capacity 8180 MW (total capacity ka 1.77%)** hai. [Q2, Q3]

3. Thorium — India Self-Sufficient (Not Uranium)

- India Thorium (monazite se extract hota hai) ki supply mein self-sufficient hai — Uranium ke liye import par nirbhar hai. [Q4]

4. Plant-State Match (Rawatbhata, Tarapur, Narora, Kakrapara)

- Match: Rawatbhata (Kota ke paas) — Rajasthan, Tarapur — Maharashtra, Narora — Uttar Pradesh, Kakrapara — Gujarat. [Q5]

5. Kaiga — Karnataka (Not Gujarat/Rajasthan)

- State-plant match: Gujarat — Kakrapara, Karnataka — Kaiga, Rajasthan — Rawatbhata, UP — Narora. [Q6]

6. Commissioning Year Match — All Disputed

- Actual commissioning years: **Kota — 1973, Kakrapara — 1993, Kaiga — 2000, Kalpakkam — 1984** — diye gaye question mein saare match galat the, isliye poora question disputed hai. [Q7]

⚠ EXAM TRAP — Commissioning years ka match galat diya gaya tha — Kota-1973, Kakrapara-1993, Kaiga-2000, Kalpakkam-1984 hi sahi hain

7. Nuclear Energy Potential (Assertion-Reason)

- Nuclear energy India ke liye futuristic demand ka promising source hai (Assertion sahi) — **lekin nuclear minerals India mein 'ubiquitously' (har jagah) available NAHI hain (Reason galat)** — Uranium kam quantity mein hai (import-dependent), Thorium abundant hai par sab jagah nahi milta. [Q8]

8. Heavy Water Plant Match (Thal, Manuguru, Kakrapara, Kaiga)

- Match: **Thal — Maharashtra, Manuguru — Andhra Pradesh (ab Telangana), Kakrapara — Gujarat, Kaiga — Karnataka.** [Q9]

9. Ennore — Tamil Nadu (Not Meghalaya)

- Diye gaye set mein 'Ennore-Meghalaya' **GALAT** hai (Ennore asal mein Tamil Nadu mein hai) — baaki options sahi hain. [Q10]

⚠ EXAM TRAP — Ennore — Tamil Nadu mein hai, Meghalaya mein NAHI

10. Kudankulam — Tamil Nadu (Map-Based Question)

- Kudankulam Nuclear Power Plant (2×1000 MW) **Kattabomman, Tirunelveli district, Tamil Nadu** mein hai. [Q11, Q12]

11. Kudankulam — Russia's 6-Unit Agreement

- 1988 mein Mikhail Gorbachev (Soviet head) ne **2 reactors** ke liye Inter-Government Agreement sign kiya tha — baad mein 6 units tak expand hua. [Q13]

12. 25th Nuclear Plant — Rawatbhata (Rajasthan)

- India ka **25th Nuclear Plant Rawatbhata (Rajasthan)** mein bana — construction July 2011 mein shuru hua. [Q14]

13. 20th Nuclear Station — Kaiga (Karnataka)

- India ka **20th nuclear power station Kaiga, Karnataka** mein hai. [Q15]

14. Heavy Water Plants — Kalpakkam Excluded

- Heavy water plants: **Hazira, Vadodara (Gujarat), Kota (Rajasthan), Manuguru (Telangana), Talcher (Odisha), Thal (Maharashtra), Tuticorin (TN)** — **Kalpakkam** mein heavy-water plant **NAHI** hai (wahan nuclear **POWER** station hai). [Q16]

15. Mithivirdi — USA Collaboration (Gujarat)

- Mithivirdi Nuclear Plant (Talaja Taluka, Bhavnagar, Gujarat) USA ke collaboration se banega — 2008 ke Indo-US nuclear deal ka hissa. [Q17]

16. Anushakti Vidyut Nigam — NPCIL+NTPC Joint Venture

- Anushakti Vidyut Nigam Limited NPCIL (51%) aur NTPC (49%) ka joint venture hai — 27 January 2011 ko incorporate hua. [Q18]

17. Haripur — West Bengal (Not Haryana)

- Jaitpur-Maharashtra, Mithi Viridi-Gujarat, Kovvadda-Andhra Pradesh sahi hain — 'Haripur-Haryana' GALAT hai (Haripur asal mein West Bengal mein hai). [Q19]

⚠ EXAM TRAP — Haripur — West Bengal mein hai, Haryana mein NAHI (naam-samanta ka trap)

18. Kalapakkam — Tamil Nadu (Not Karnataka)

- Kakrapar-Gujarat, Kudankulam-Tamil Nadu sahi hain — 'Kaiga-Uttar Pradesh' aur 'Kalapakkam-Karnataka' dono GALAT hain (Kaiga Karnataka mein hai, Kalapakkam Tamil Nadu mein hai) — is question mein 2 galat combinations the, isliye disputed hai. [Q20]

19. Largest Installed Capacity — Tarapur (Historical Answer)

- Diye gaye options mein installed capacity: **Kakrapar 1840 MW, Kaiga 880 MW, Narora 440 MW, Tarapur 1400 MW** — historical context mein Tarapur ko sahi answer maana gaya (recent data mein Kakrapar ki capacity zyada badh gayi hai). [Q21]